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दीजिए

Day 15

Total Sales = Selling Price $\times$ Units Sold

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	X		Y		Z	
	Unit price (₹/unit)	Sales (₹ lakhs)	Unit Price (₹/unit)	Sales (lacs)	Unit price (₹/unit)	Sales (₹ lakhs)
2011	5	3	4	4.4	7	3.5
2012	6	3	6	4.2	6	3.6
2013	7	3.5	5	4.5	8	4
2014	8	4	7	4.9	5	5
2015	9	3.6	6	4.2	6	4.2

(a) Price of Soap by X in 2013 = ₹ 7/unit

Price of Soap by Y in 2013 = ₹ 5/unit

Price of Soap by Z in 2013 = ₹ 8/unit

$$\text{Average price} = \frac{7+5+8}{3} = \frac{20}{3} = ₹ 6.67/\text{unit}$$

$\therefore$ the average price of Soap in 2013 = ₹ 6.67/unit

(b) No. of units sold for X in 2012

$$\Rightarrow \frac{3 \times 100000}{6} = 50000 \quad \text{--- (i)}$$

No. of units sold for X in 2011

$$\Rightarrow \frac{3 \times 100000}{5} = 60000 \quad \text{--- (ii)}$$

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from (i) & (ii), we can find decrease in units sold

$$\Rightarrow 60000 - 50000 = 10000$$

$$\text{Required Percentage decrease} = \frac{10000}{60000} \times 100$$

$$= 16.66\%$$

$\therefore$ the percentage decrease in units sold for X in 2012 over 2011 is 16.66%

(c) Number of units sold for X in 2013

$$\Rightarrow \frac{3.5}{7} \times 100000 = 50000 \quad \text{--- (i)}$$

Number of units sold for Y in 2013

$$\Rightarrow \frac{4.5}{5} \times 100000 = 90000 \quad \text{--- (ii)}$$

Number of units sold for Z in 2013

$$\Rightarrow \frac{4}{8} \times 100000 = 50000 \quad \text{--- (iii)}$$

Total units sold in 2013 = (i) + (ii) + (iii)

$$\Rightarrow 50000 + 90000 + 50000 = 190000$$

$$\text{Percentage share of Z} = \frac{50000}{190000} \times 100$$

$$= 26.31\%$$

$\therefore$ the percentage share of Z in the total units sold in 2013 = 26.31%

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(d) No. of units sold for Y in 2011

$$\Rightarrow \frac{4.4}{4} \times 100000 = 110000 - \textcircled{i}$$

No. of units sold for Y in 2012 $\Rightarrow \frac{4.3}{6} \times 100000$

$$= 70000 - \textcircled{ii}$$

No. of units sold for Y in 2013 $\Rightarrow \frac{4.5}{5} \times 100000$

$$= 90000 - \textcircled{iii}$$

No. of units sold for Y in 2014 $\Rightarrow \frac{4.9}{7} \times 100000$

$$= 70000 - \textcircled{iv}$$

No. of units sold for Y in 2015 $\Rightarrow \frac{4.2}{6} \times 100000$

$$= 70000 - \textcircled{v}$$

from equation $\textcircled{i}$, $\textcircled{ii}$, $\textcircled{iii}$, $\textcircled{iv}$ & $\textcircled{v}$

Y sold the maximum number of units in 2011

(e) Maximum price per unit given in the table = ₹ 9

Minimum price per unit given in the table = ₹ 4

$$\text{Required percentage} = \frac{9-4}{4} \times 100 = \frac{5}{4} \times 100 = 125\%$$

$\therefore$ the maximum price per unit exceeds the minimum price per unit by 125%